

Enterprise architecture in an agile world

PwC's point of view



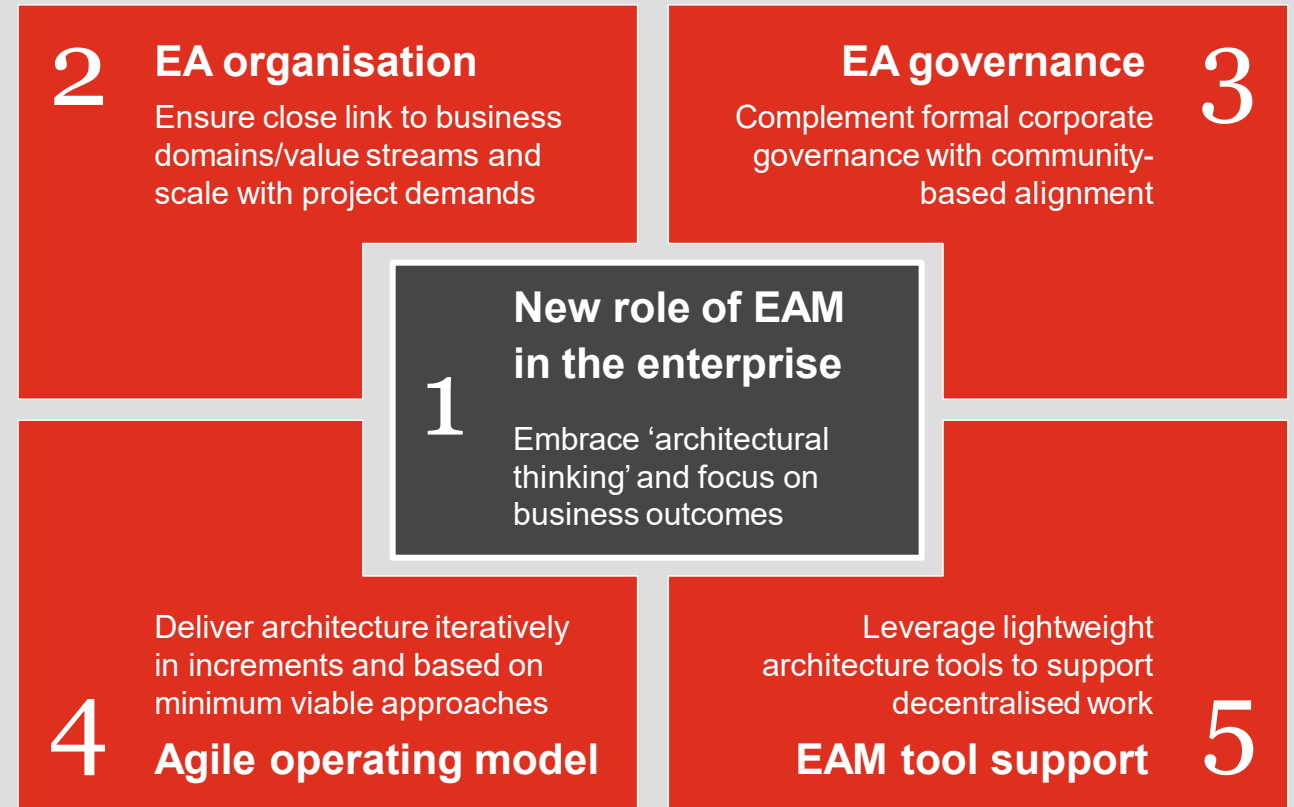
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In today's increasingly connected world, there is tremendous pressure on organisations to lead digital transformations by adopting new technologies, scaling rapidly, and reducing cost. Executing digital transformations requires organisations to use more agile ways to build complex systems.

The question that is often asked is “**how does enterprise architecture management (EAM) fit with agile approaches?**”. Many people consider enterprise architecture and agile methods to be two conflicting concepts.

In this paper, we elaborate on five areas of change which we think are crucial for enterprise architecture functions to enable agile organisations.

The enterprise architecture (EA) function needs to adapt in five core areas to effectively support agile enterprises.





EA organisation

Ensure close link to business domains/value streams and scale with project demands

EA governance

Complement formal corporate governance with community-based alignment

New role of EAM in the enterprise

Embrace 'architectural thinking' and focus on business outcomes

Deliver architecture iteratively in increments and based on minimum viable approaches

Agile operating model

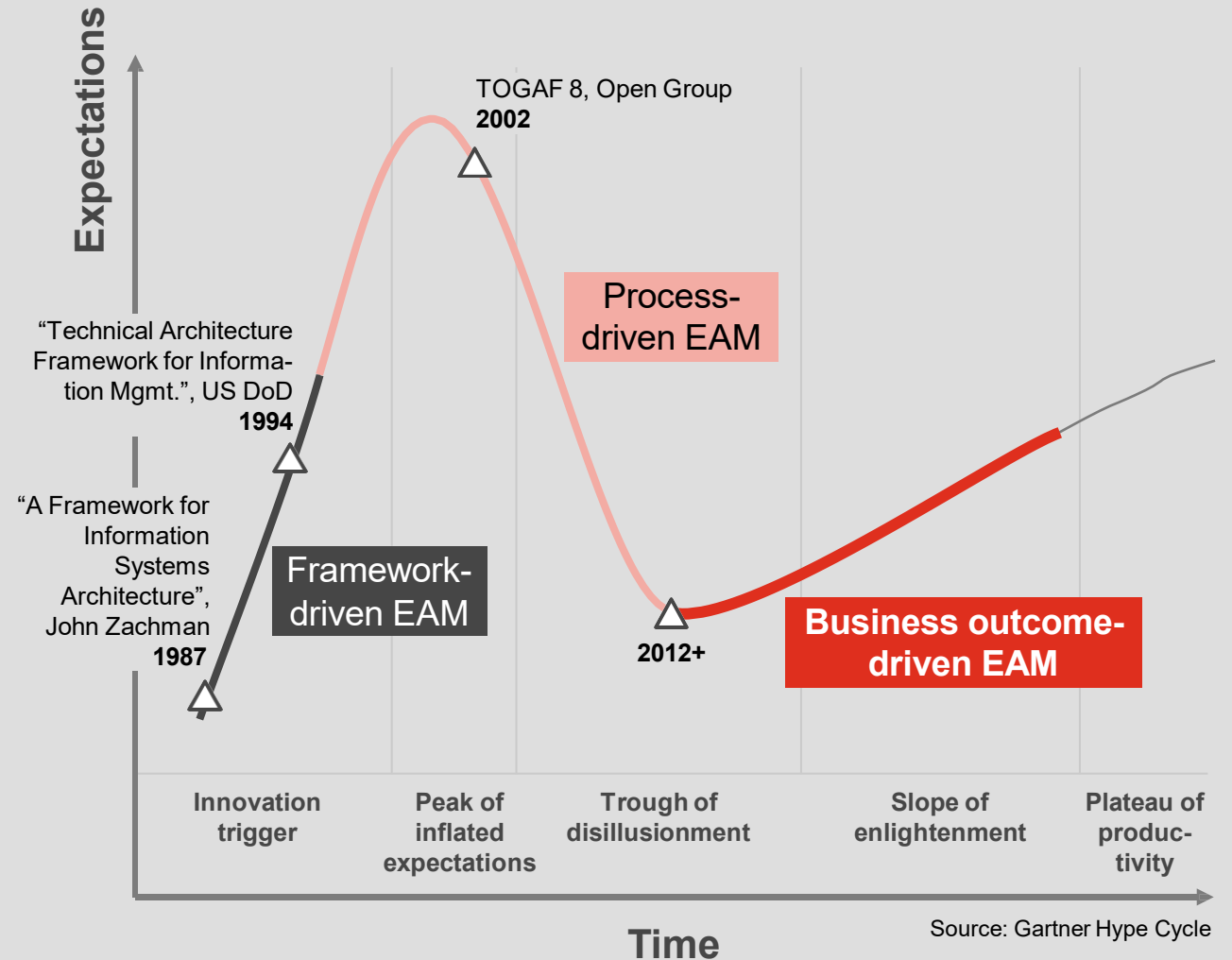
Leverage lightweight architecture tools to support decentralised work

EAM tool support

We are seeing a general trend towards business outcome-oriented EAM capabilities

The enterprise architecture management (EAM) discipline has developed significantly over recent years. The Gartner Hype Cycle shows that EAM is currently in the 'slope of enlightenment'.

The focus of EAM has shifted away from its former internal orientation on EAM frameworks and processes, and more towards the business's needs. Modern EAM capabilities are driven by achieving measurable business outcomes and making real-world impacts.



The introduction of agile working poses further challenges for an effective enterprise architecture function

New
challenges of
agile working

Existing
challenges for
the EA function

Agile working approaches change the way IT solutions are developed and delivered. This creates **new challenges** for the EA function ...

- + **Solution architectures** are not known at the start of the project; they **evolve continuously** instead
- + Delivery cycles (sprints) are shortened to weeks – instead of months – requiring **fast architecture feedback** and constant involvement
- + **High autonomy of development teams** conflicts with formal corporate governance
- + **Loss of waterfall-based stage gates** for architecture involvement

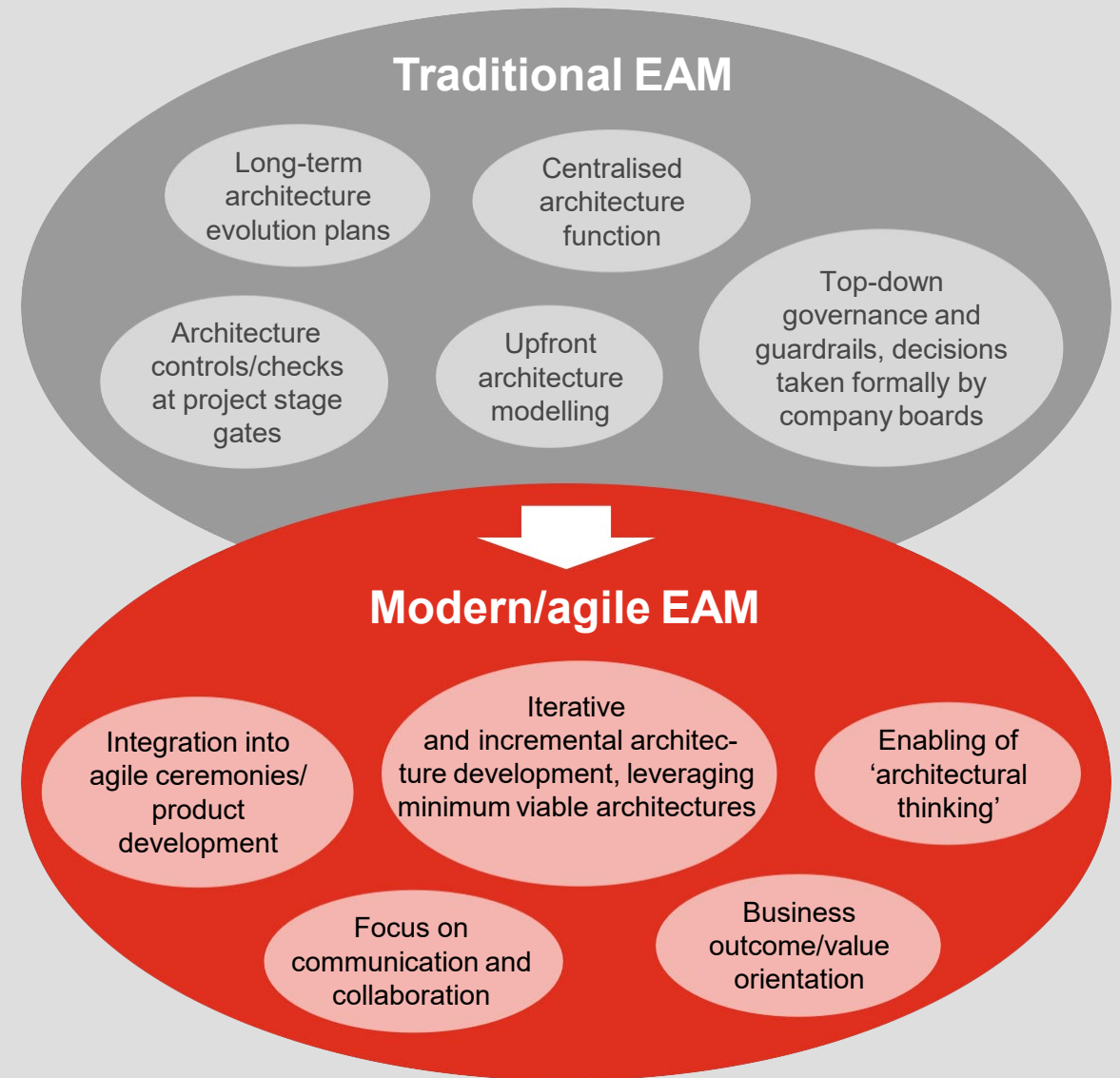
In many organisations, we have seen deficits in how the enterprise architecture function delivers business value and how it is integrated with the various stakeholders. You may have noticed these in your enterprise as well ...

- × Lack of understanding and support from business management
- × Benefits or ROI of the EA work are hard to measure
- × Architects seen as objectors, hindering project progress and being inflexible
- × EA not involved in core decisions, or is even circumvented
- × Target architectures and frameworks that are developed are not known and do not get used or implemented in practical work
- × The architecture repository is outdated and only used by a few architects

The future role of EA will shift to establish 'architectural thinking' within the enterprise

A modern architecture approach still utilises the tools of traditional EAM, such as governance, architecture modelling and planning. However, we are seeing that successful EA functions have developed the following new capabilities, which have become key to their success:

- Alignment of IT and business, with clear orientation towards business value
- Iterative and incremental development of architectures
- Integration into agile delivery organisations
- Enabling of 'architectural thinking' among teams and in the organisation as a whole
- Balance of top-down governance with bottom-up community approaches



In modern EAM, architects act as strategic advisors to the business to create deeper understanding of the impact of technology on strategy and execution



**Enterprise
architecture
and
technology**



**Digital
business
strategy**

Scenario planning and recommendations. Analysis of emerging technologies, business models, trends and innovations



**Enterprise
change**

Architecture analysis of current and future business capabilities (people, processes, information, technology) and recommendations for IT priorities



**Technology
and
innovation**

Risk analysis of business models and technology decisions. Advice on use of guidelines and standards. Facilitation of experiments and proof of concepts

Understanding of the impact of emerging technologies on the business and operating models

Roadmap showing how to close the gap between current and future business capabilities

Awareness of where to be flexible in innovating and experimenting



Business



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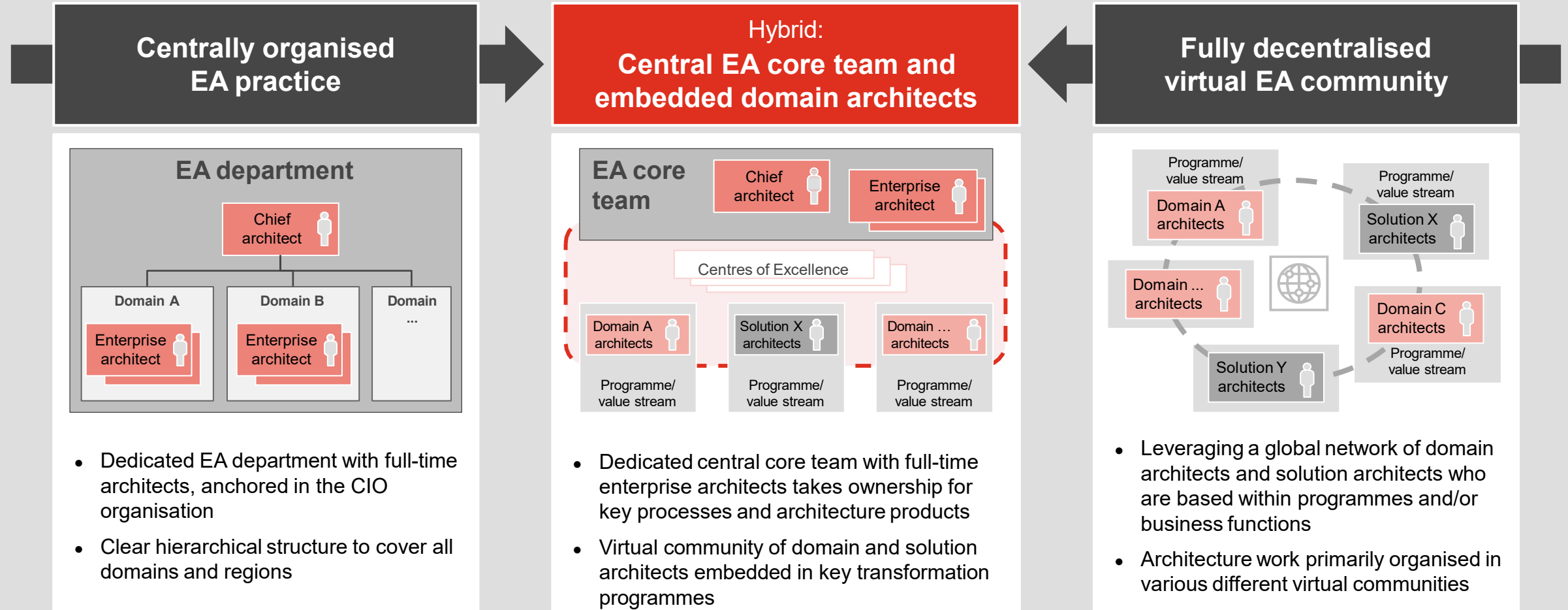
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EAM tool support

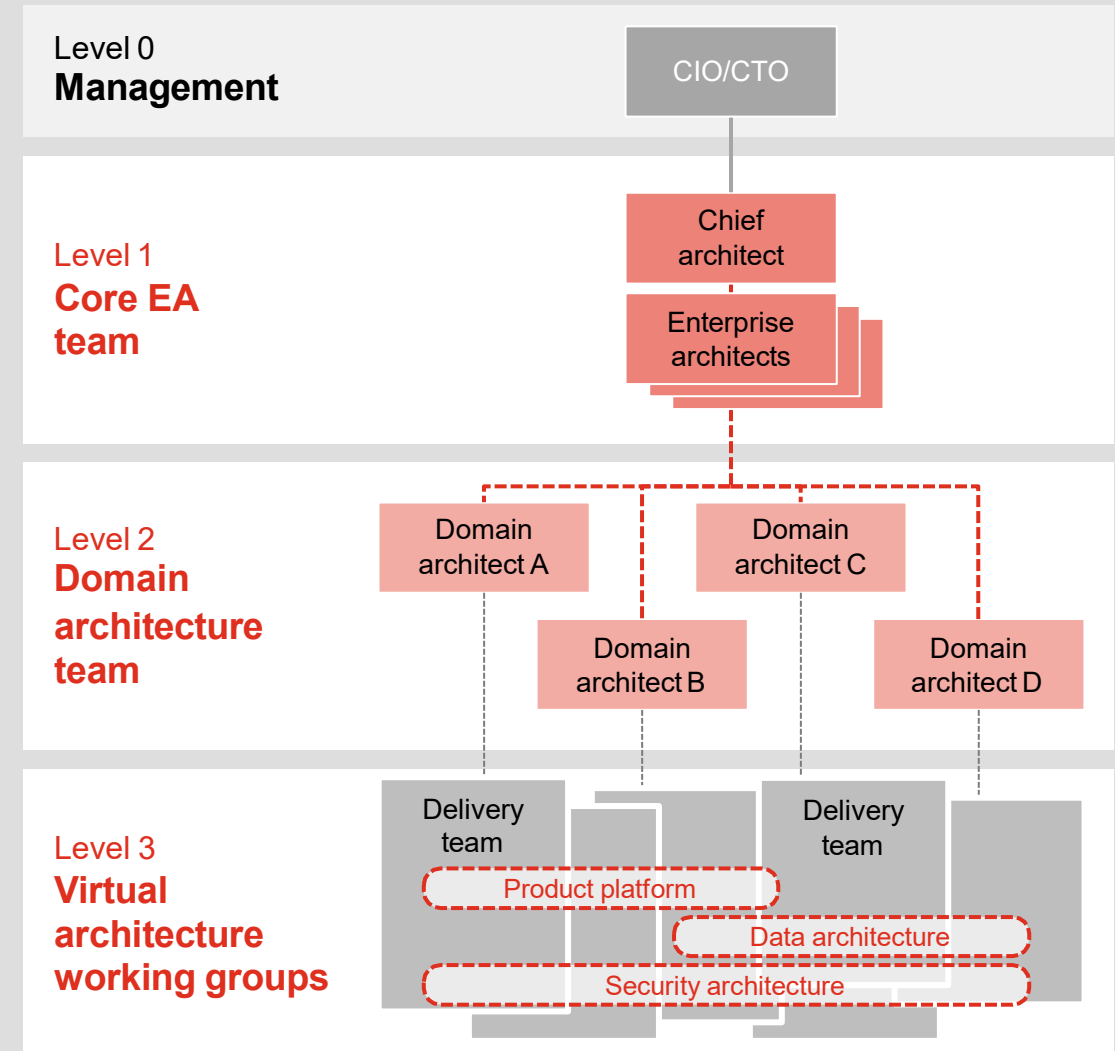
One potential future organisational setup of EA combines a central core team with embedded domain architects



A hybrid EA organisation ensures a close link to business demands

The proposed hybrid organisational structure is a federated model consisting of:

1. **A lean, central core EA team** for high-level EA strategy development and support of enterprise planning processes. Its core tasks are to develop an architecture vision, align architectural strategy with business needs, drive innovation – with a focus on value for the business – and define company-wide architecture standards and guidelines.
2. **Decentralised, integrated domain architects** for business-oriented development of domain and solution architectures for specific business or project needs. These architect roles are focused on specific business domains (e.g. customer interaction, sales, finance). Their key duties are to develop domain-specific architectures and continuously align with projects/agile teams to support them in creating solution architectures. They coach and guide the teams on ‘architectural thinking’. They foster reuse of architecture assets and knowledge sharing between teams.
3. **Virtual architecture working groups** to leverage exchange of information and best practice between enterprise, domain and solution architects (e.g. in the form of an architecture chapter) and to integrate with non-architecture communities.



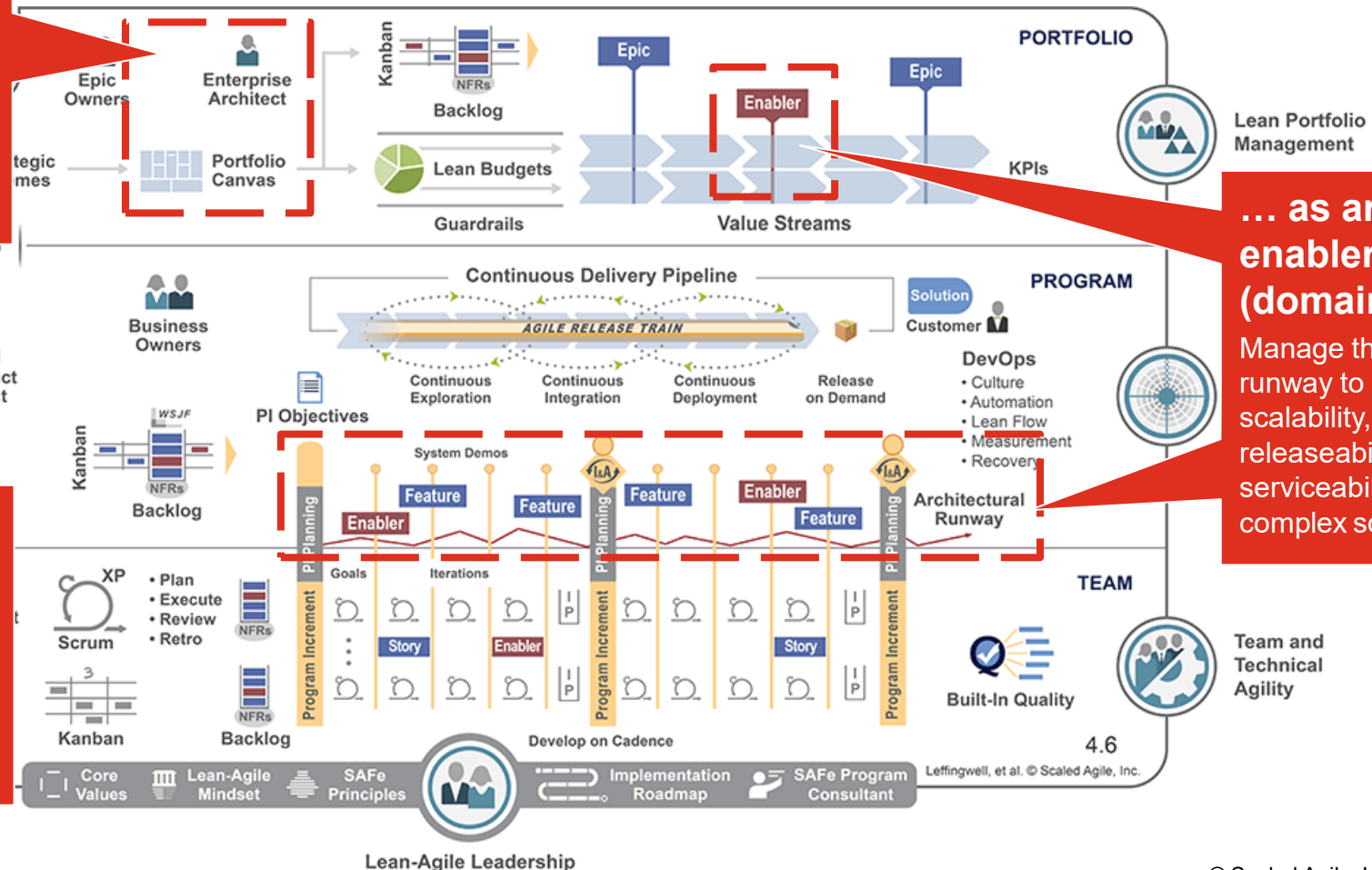
Architects have different roles within agile enterprise frameworks (example: SAFe®)

... as strategic planners (enterprise architects)

Drive architectural initiatives for the business portfolio. Work across Solution Trains and Agile Release Trains to define key technical initiatives.

... as lean-agile leaders (solution architects)

Define and communicate a shared technical and architectural vision for and across Agile Release Trains. Promote reuse of ideas, components, services and proven patterns.



... as architecture enablers (domain architects)

Manage the architectural runway to ensure scalability, modularity, releaseability and serviceability in large and complex solutions.



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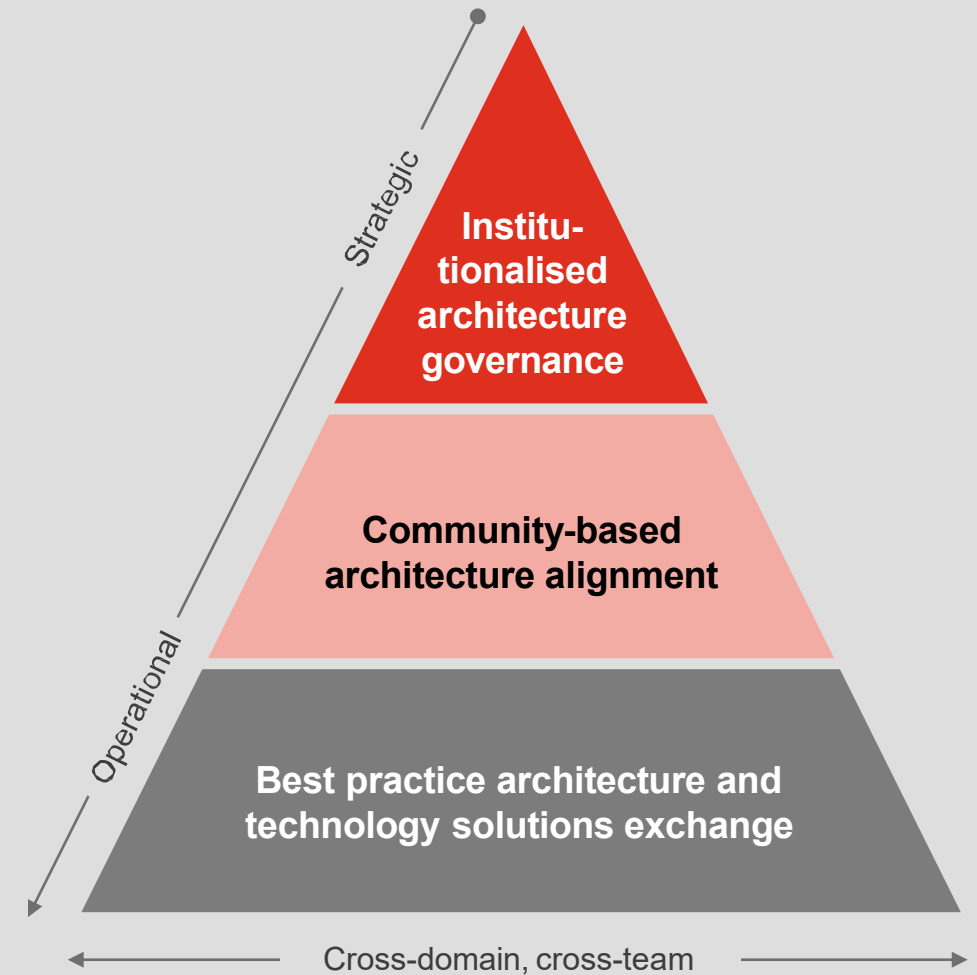
EAM tool support

EA governance will partly rely on corporate bodies, but needs to be complemented with community-based structures

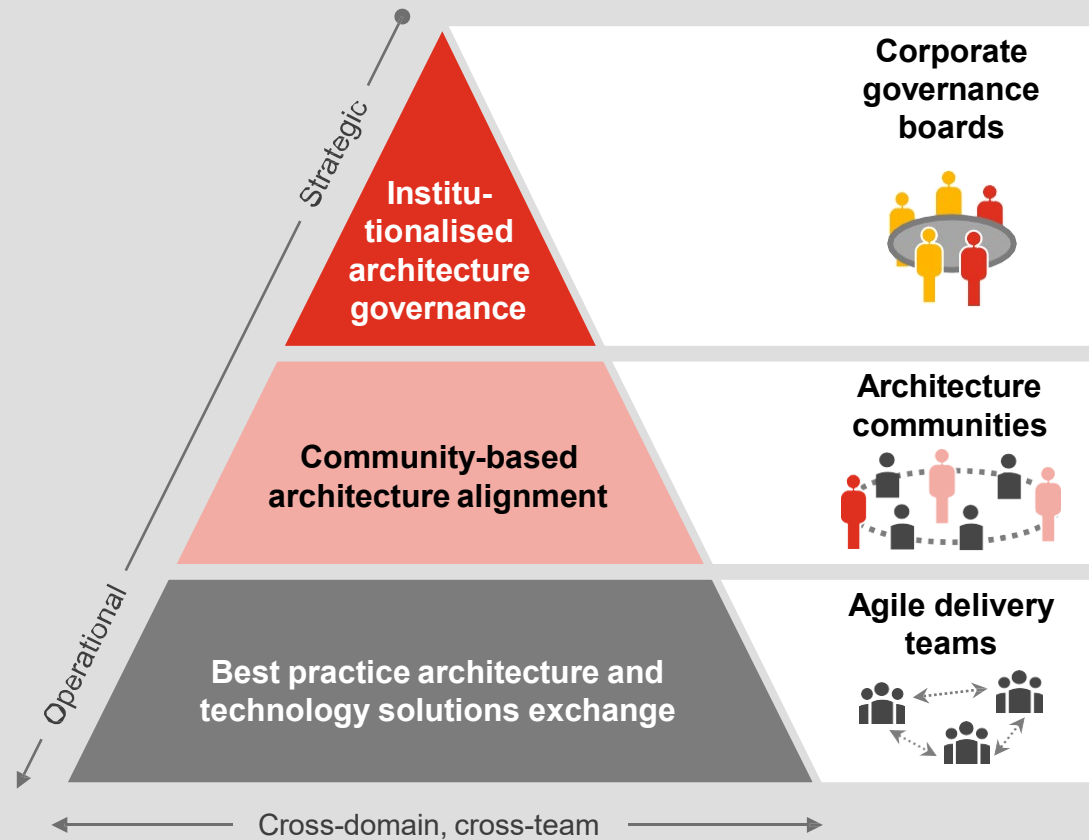
The rationale of this layered governance model is that **strategic architecture decisions** still need to be well aligned with the overall business and digitalisation strategies. Therefore, these decision issues should be formally presented, discussed and approved by the relevant corporate architecture governance boards.

Tactical architecture matters need to be aligned across different business domains and teams. But this alignment can be self-organised within the architecture community, without the need for time-consuming decisions by formal governance bodies.

Lastly, we believe that delivery teams need certain freedom – within given architectural limits – to come up with the best-suited, **practice-based solutions**. In order to do this, we have found that virtual communities should be used to foster decentralised alignment and exchange of best practice across teams. This also generates valuable real-world feedback to inform architecture standardisation and technology decisions at the higher governance levels.



Informal architecture governance/alignment leverages collaboration with agile delivery teams



Example Issues

- Strategic direction setting (e.g. strategic themes)
- Business alignment and business value
- (Intentional) business architecture and capability development
- Innovation strategy and roadmap

Governance rationale

Strategic architecture decisions are aligned with business goals and drive business value (connecting business and IT stakeholders). Enterprise architects ensure alignment of strategy and architecture.

- Architecture guidelines
- Integration strategy
- Reference architectures and patterns

Overall architecture is aligned across domains and teams. Decisions are driven by mixed architecture communities (guilds).

- De facto technology standards
- Architecture models
- (Emerging) solution architectures and designs, best practices

Loosely coupled teams allow for minimum viable standardisation based on real-world experience, with little effort required for alignment.

Delegate decisions

Delegate decisions

Real-world feedback



Business executive



Enterprise architect



Domain architect



Delivery team incl. solutions architect



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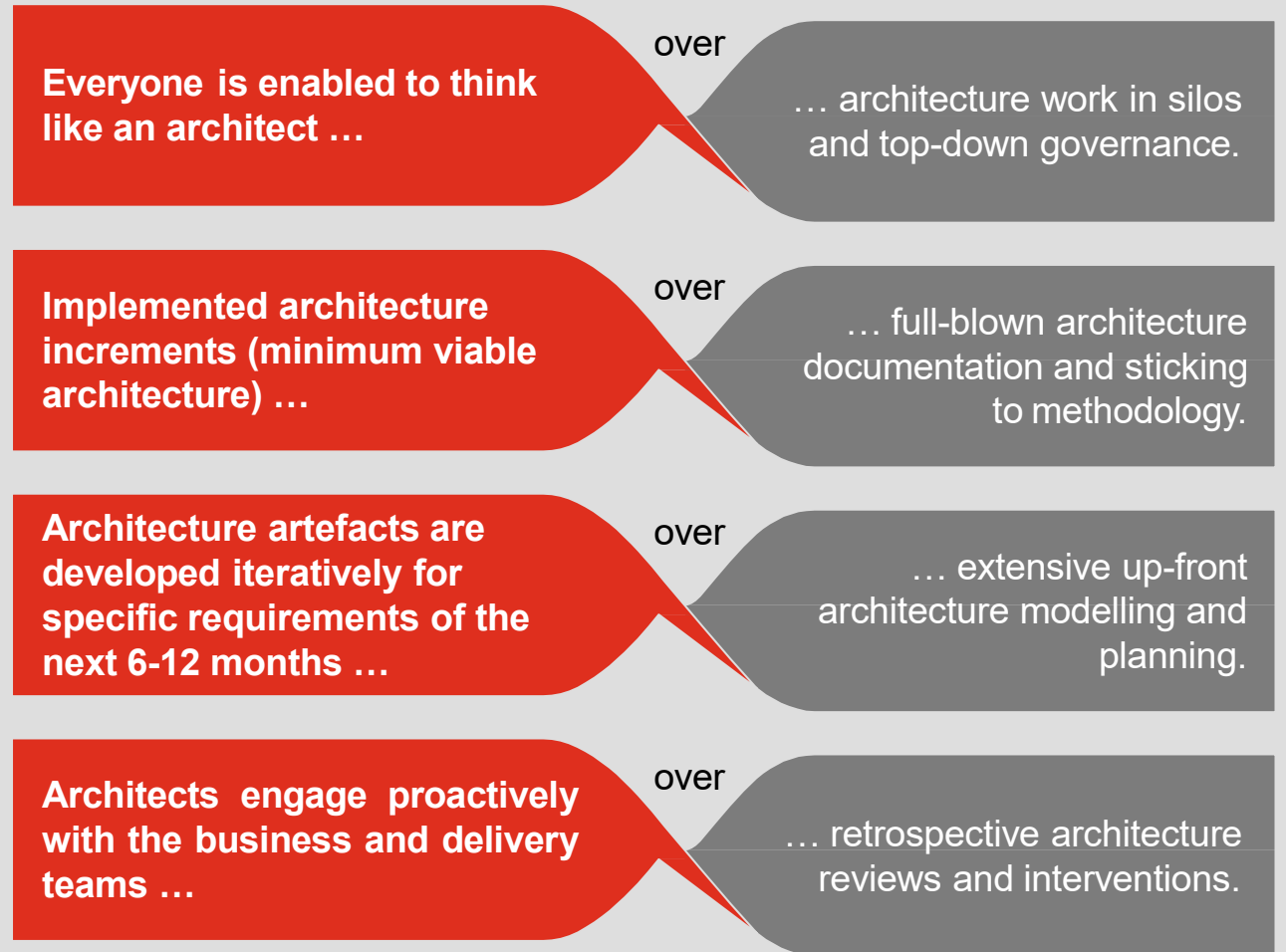
EAM tool support

Future EA functions need to adopt agile and lean principles

A modern EA operating model should revolve around some key principles:

- **Pragmatic and value-oriented:** EA efforts focused on key stakeholders, not methodology or technology.
- **Lean:** keep it simple with fit-for-purpose architecture. Avoid architectural 'waste' and enable flow of work.
- **Agile:** work iteratively and incrementally, starting with artefacts adding most business value. Embrace change and quickly adapt to new requirements.
- **Collaborative:** work closely with the business and delivery teams to include their feedback, provide governance to streamline development and ensure alignment with the business strategy.

An agile EAM manifesto can create a framework for a renewed way of architectural working



The delivery cycles of EA services have shortened significantly in agile environments

Strategy and guidelines

- Develop architecture vision aligned to business objectives and customer value
- Guide creation of fundamental architecture concepts
- Provide architectural guidelines and direction for cross-programme alignment
- Utilise training and communication to bring architecture to the delivery teams

Technology standards

- Facilitate selection of technology standards and products across domains and teams
- Actively collaborate with delivery teams to evolve technology standards based on their real-world needs and experiences
- Maintain and evolve technology roadmaps

Innovation management

- Perform top-down technology innovation research to develop trend radars
- Probe technological innovation from the bottom up with proofs of concept and rapid prototyping
- Exchange experiences/best practice across teams within communities of interest (bottom-up)

Architecture planning

- Develop flexible business domain architectures
- Link architecture features and epics to business capabilities to ensure prioritisation of business value
- Maintain architectural runway, identify minimum viable architectures

Architecture consulting

- Coach and mentor teams in architectural issues and establish 'architectural thinking'
- Promote cross-programme alignment to foster reuse of architecture assets across delivery teams
- Provide guidance on architecture principles and technology standards

Architecture documentation

- Develop fit-for-purpose architecture artefacts in increments, continuously including the customer's feedback
- Guide development and usage of light-weight, pragmatic architecture models and patterns

x months

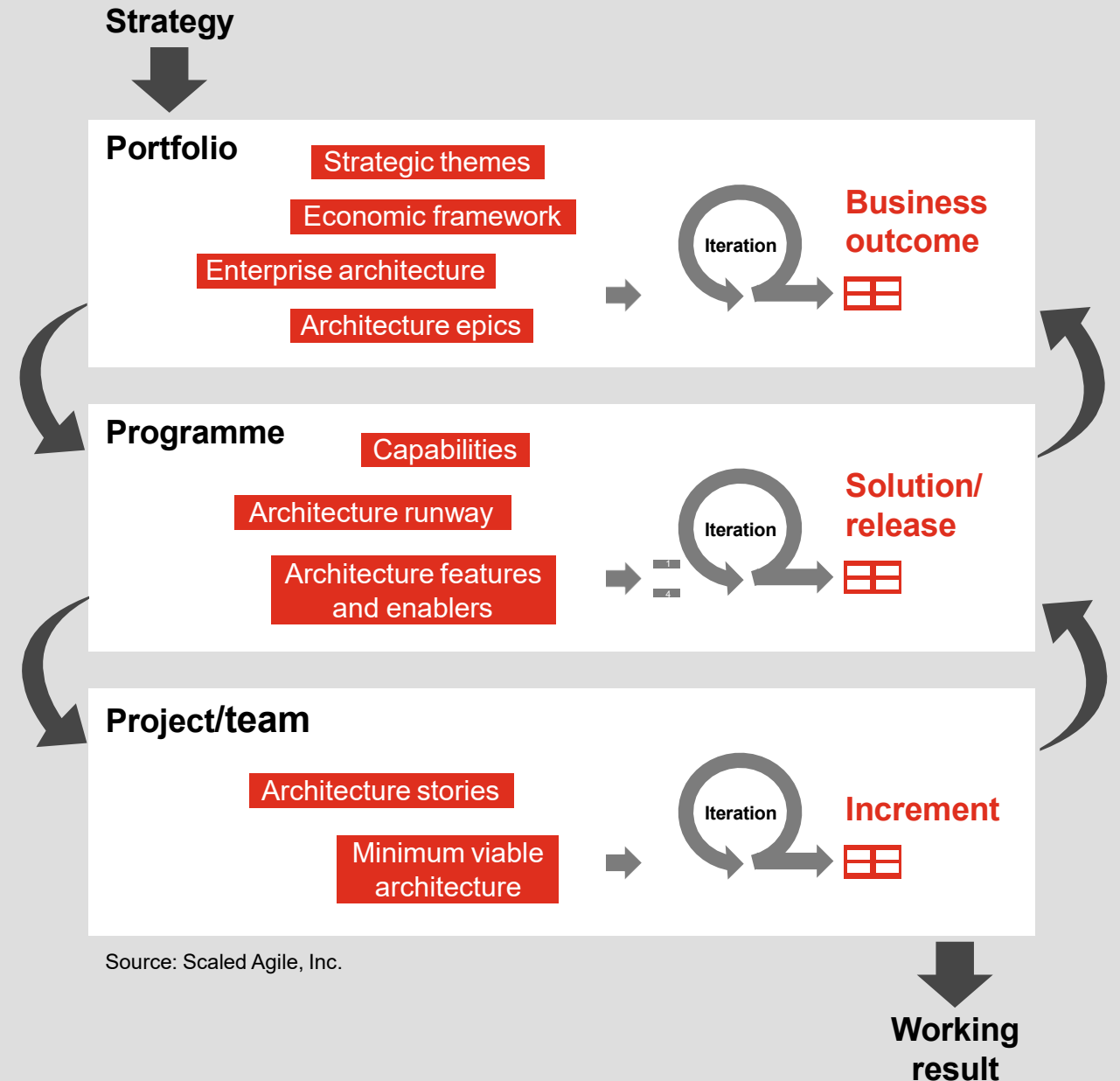
1-2 years

2+ years

Architecture work needs to happen on different corporate levels

Architecture teams following a lean-agile operating model are engaged in various activities:

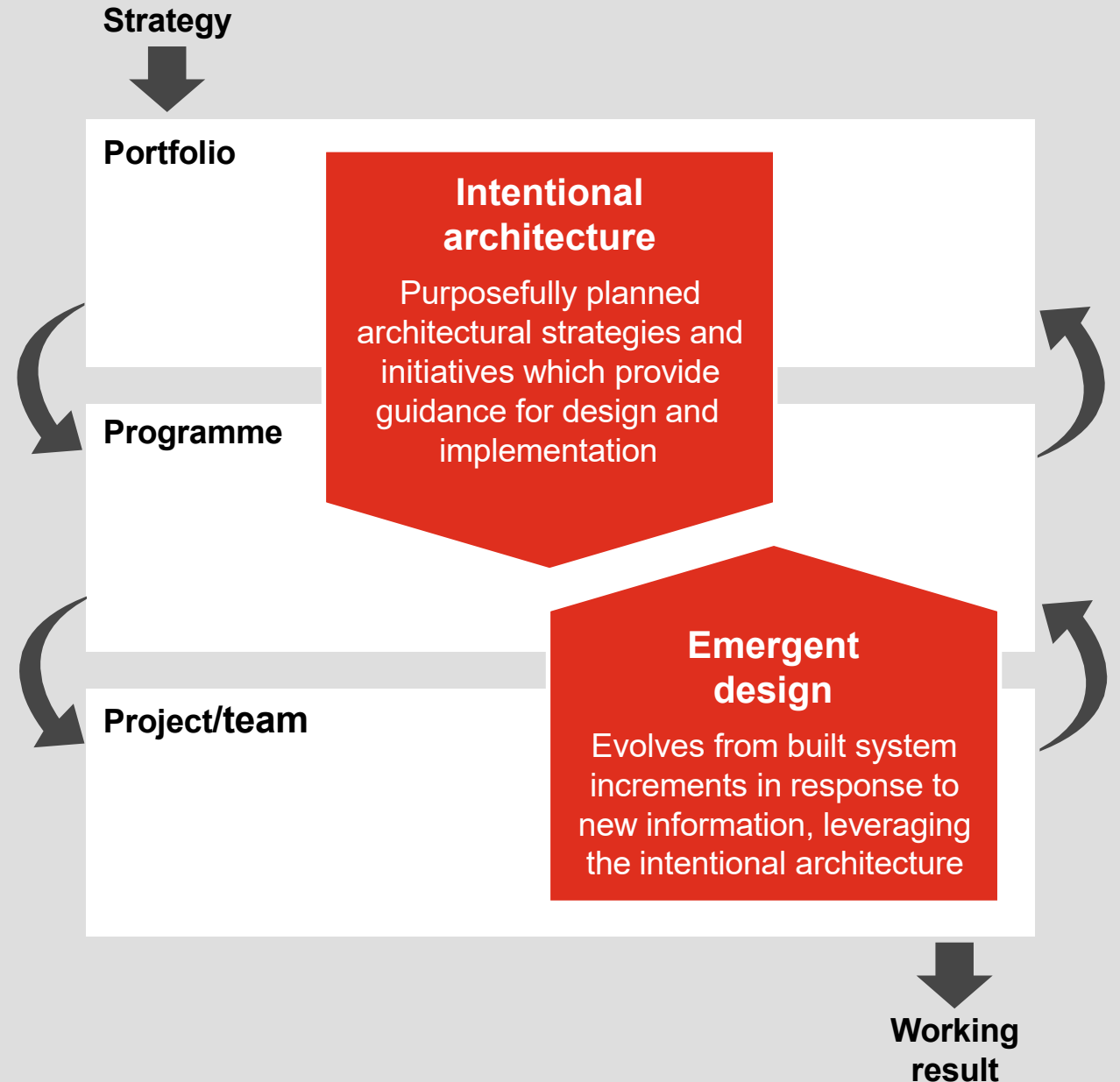
- On the **portfolio level**, enterprise architects contribute to business/IT alignment. They support end-to-end value streams by translating the enterprise strategy into relevant strategic themes and architecture epics.
- Across different **programmes/solutions**, the domain architects plan and refine the architectural runway for the delivery teams. This is done jointly with product management to ensure fit-for-purpose architecture solutions.
- In collaboration with **project teams**, the solution architects define stories to incrementally build up minimum viable architectures. They contribute best practice, lessons learnt and standards defined at the portfolio or programme level.



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While we must acknowledge emergence in design and system development, a little planning can avoid much waste.

James Coplien, Lean Architecture



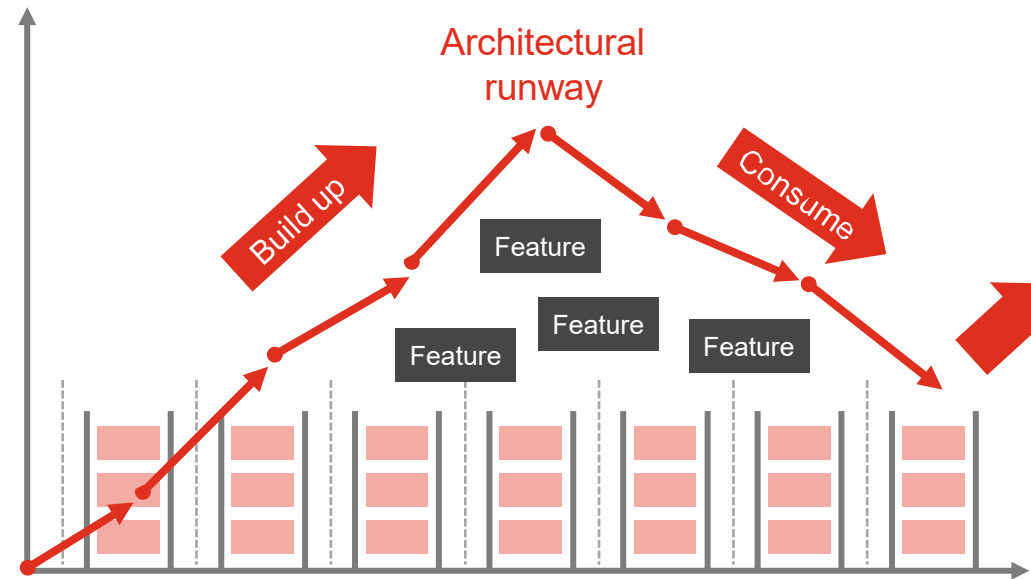
Architecture development needs to be naturally incorporated into the agile implementation process

Architects work on various levels to build up an architectural runway for the delivery teams:

- Enterprise architects define architecture enablers at portfolio level to realise the architecture strategy and respond to strategic business demands.
- Domain/solution architects refine these enablers within their programmes and guide the implementation teams (e.g. using agile planning tools).

Example: architectural runway in SAFe®

Technical foundation to support near-term feature implementation



Source: Scaled Agile, Inc.

Investment in the architecture is guided by enablers, which address technical debt or capabilities required by feature teams:

- There is a commitment to invest in these enablers incrementally – based on specific demand – but continuously, e.g. by reserving fixed capacity (story points) in each sprint/increment.
- This should avoid both excessive technical refactoring and large amounts of effort in up-front architecture design.

5

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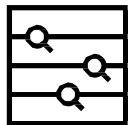
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Lightweight EAM tools support value-oriented and decentralised architecture work

A comprehensive EA repository demonstrates the value of enterprise architecture by ...



Providing an accurate and up-to-date view of business capabilities, applications, data and technology



Supporting architecture reports which can be directly used for decisions or in management presentations



Being accessible to anybody in the company, allowing them to see what the company's architecture looks like

However, we often find that established architecture repositories are outdated or only used by the architects. In many cases, significant investment was made in the initial build-up of these databases. But due to a lack of engagement with the application owners or projects, these data assets are not updated or leveraged in the day-to-day work.

Based on our experience, **successful modern EAM tools stand out due to ...**



Easy deployment and configuration – ideally cloud-based, focussing on business value instead of tool implementation



Intuitive user interface – enabling involvement of all users, not just architects



Openness and collaboration/workflows – fostering engagement between IT and business users and integrating other enterprise data assets

Case study: EAM tool implementation in an agile environment

Client challenge

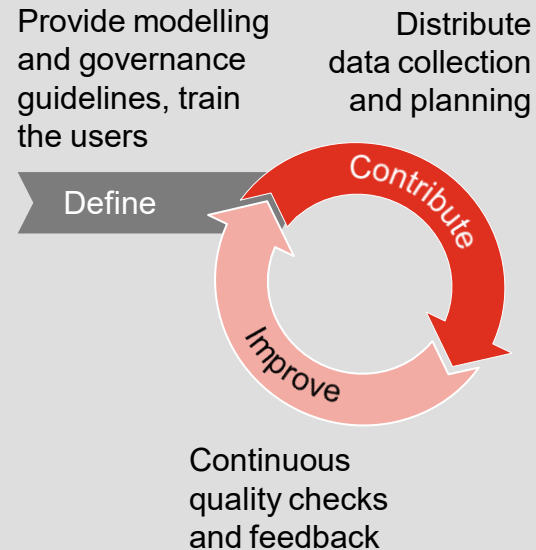
- Agile setup of EA led to architects forming a new image of themselves. This had implications on their EAM tool requirements

Tool approach

- The new agile setup required everybody in the IT organisation to be able to access the tool
- EAM reports had to be created by new stakeholders – business as well as IT users
- Information had to be maintained directly by the relevant owners – application managers, process owners, data stewards etc.

Benefits

- The value of the architects' work became more visible
- Everybody benefitted from the knowledge about available technologies and inter-dependencies
- Blind spots in shadow IT were revealed



Centralised

The new objectives had a big impact on **tool selection**, as only few of the IT personnel had the skills required to use the established EA tool suites.

Modelling guidelines had to be easily understandable. Sometimes, EA modelling was simplified at the expense of detail to motivate more people to enter information.

A **governance framework** defined responsibilities so that it was clear who had to maintain what kind of information.

Distributed

Training sessions ensured that data was properly collected and involved a wide range of users. They were tailored for different target groups, such as business users and IT teams.

Data collection happened in a de-centralised manner. Data completion reports stimulated contributors from various different areas to provide as much information as possible.

Due to the distributed nature of data entry, **data quality** had to be checked constantly, and the architects invested in data correction and re-training of users.

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